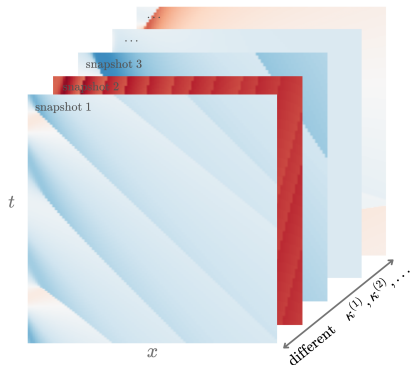


TEMPO: Regime-Aware Operator Learning with Locally Adapted Bases

$$D = \{ (u^{(i)}, \kappa^{(i)}, t^{(i)}, s^{(i)}) \}_{i=1}^N$$



Offline: EM-GMM $\Rightarrow M$ bases $\{\Phi_m\}$

$$\text{Online: } \hat{s} = \sum_m w_m \sum_k b_{mk} \Phi_k^{(m)}$$

Operator learning problem:

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \rightarrow L^2(\Omega \times \mathcal{T}), \quad (u, \kappa) \mapsto s$$

Learn $\hat{\mathcal{G}} \approx \mathcal{G}$ from data $\mathcal{D}$ without solving the PDE.

Challenge

A single basis fails when solutions change character fundamentally across κ .

TEMPO

Discover M latent regimes via EM-GMM, build a dedicated basis $\{\Phi_m\}$ per regime (*POD* or *NeuralPOD*), route new inputs at inference time.

$\Phi_k^{(m)}$	POD mode k of regime m
$w_m(u, \kappa)$	gating weight
γ_{im}	soft EM assignment